

A123 ANR26650M1B LFP Cell Physical Modeling and Calibration Using PyBaMM

1. Introduction

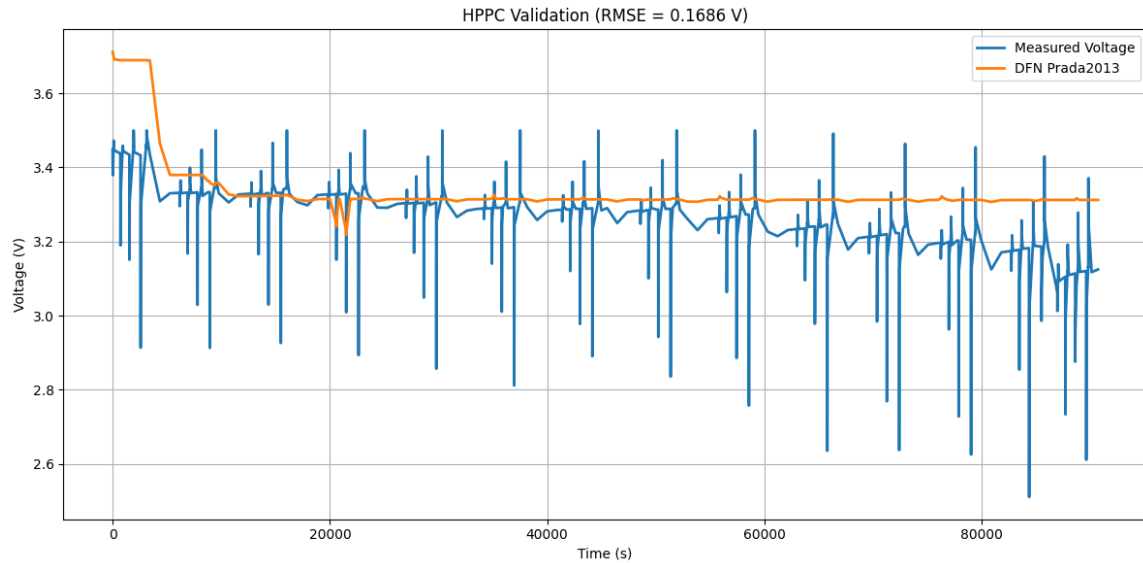
This study develops, calibrates, and validates a DFN electrochemical model for the A123 ANR26650M1B LFP/Graphite cell using PyBaMM. HPPC and discharge datasets were used for calibration and validation.

2. Methodology

A DFN model with the Prada2013 parameter set was selected as the baseline. HPPC data were cleaned, synchronized, and applied as the model current profile. Calibration was performed sequentially to maintain physical interpretability.

3. Baseline DFN Simulation

The baseline DFN showed substantial voltage mismatch, capacity mismatch, and transient response errors. Baseline RMSE was approximately 0.17 V.



The above shown voltage simulation was run using the parameters of Prada2013 and current was scaled down by a factor of 10 to run the DFN code. However, there are multiple fundamental issues in this simulation – evident from the voltage profile. In the subsequent steps, I have tried to counter those issues.

4. Calibration Strategy

Calibration was performed through OCP alignment, stoichiometry optimization, geometry calibration, diffusivity tuning, kinetic optimization, electrolyte correction, and MP-DFN development.

5. OCP Calibration

A literature-derived LFP cathode OCP was implemented together with a +0.10 V alignment shift to match measured voltage behaviour.

6. Stoichiometry Optimization

Differential evolution optimization produced $\theta_{n_init} = 0.905$ and $\theta_{p_init} = 0.5075$, significantly improving discharge voltage prediction.

7. Geometry Calibration

Negative electrode thickness, positive electrode thickness, and positive active material volume fraction were adjusted to improve capacity matching.

8. Dynamic Parameter Calibration

Positive diffusivity, exchange-current multipliers, and electrolyte conductivity were calibrated to reduce transient voltage errors.

9. MP-DFN Development

An MP-DFN model including particle-size distribution was implemented to better represent heterogeneous diffusion behaviour.

10. Validation Results

The final calibrated model achieved approximately 26 mV RMSE on 1C discharge and 28–30 mV RMSE on HPPC validation.

11. Plot Analysis

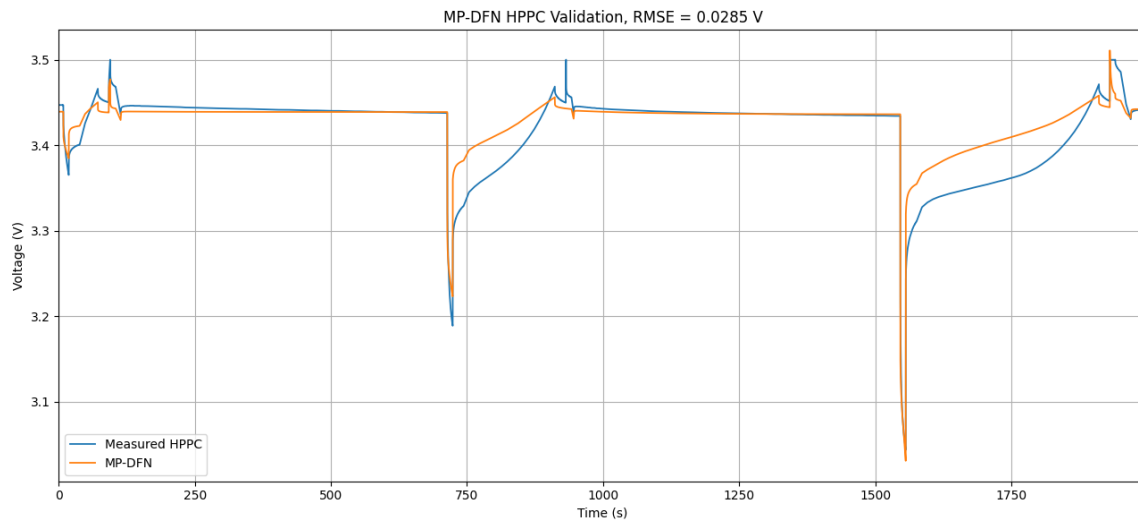
Voltage and residual plots were analysed to identify equilibrium and transient prediction errors.

12. Discussion

The final model reproduces major experimental behaviours while retaining physical meaning in calibrated parameters.

13. Conclusions

Calibration reduced model error from approximately 0.19 V to approximately 26–30 mV.



After multiple optimization strategies, I was able to optimize the DFN model. But, the only challenge that need further improvement/investigation is hysteresis nature of OCV curve. The above diagram shows the result which capture the HPPC profile current to a good extent. This plot shows the simulation vs experiment data till $t = 2000$ s. If code is utilized for the complete time interval, hysteresis issue will amplify.

Also, while plotting the charging curves of 1-4C data was analysed, an OCP offset with hysteresis was observed which confirmed my hypothesis of OCV hysteresis challenge. Although, due to submission deadline, that challenge is not fully addressed in this assignment.

Optimization Progress Summary

Stage	Parameters Modified	Result
Baseline DFN	Default Prada2013	RMSE ~ 0.19 V
Stoichiometry Optimization	θ_n , θ_p	Major improvement
Geometry Calibration	L_n , L_p , volume fraction	Capacity matched
Dynamic Calibration	Diffusivity, kinetics, conductivity	Improved HPPC response
MP-DFN	Particle-size distribution	RMSE ~ 28 mV

Final Calibrated Parameters

Parameter	Value
θ_{n_init}	0.905
θ_{p_init}	0.5075
Negative electrode thickness	129e-6 m
Positive electrode thickness	171e-6 m
Positive active material volume fraction	0.55

Positive particle radius	2.5e-6 m
Positive diffusivity	1.0e-13 m ² /s
Positive kinetic multiplier	30
Negative kinetic multiplier	30
Electrolyte conductivity multiplier	1.6
Electrolyte diffusivity multiplier	1.0
Contact resistance	0 Ω
Temperature	296.62 K
OCP shift	+0.10 V

Results Summary

Baseline DFN RMSE $\approx$ 0.19 V

Calibrated DFN RMSE $\approx$ 26–30 mV

MP-DFN RMSE $\approx$ 28 mV

Future Work

Future work includes multi-rate charge calibration, explicit LFP hysteresis modelling, thermal coupling, PSD validation, and long-duration HPPC validation.

Reference

1. OCV plot references – [Paper 1](#) & [Paper 2](#)
2. Physical parameters like particle radii, electrodes thickness were taken from actual data available for LithiumWerks LFP cell.